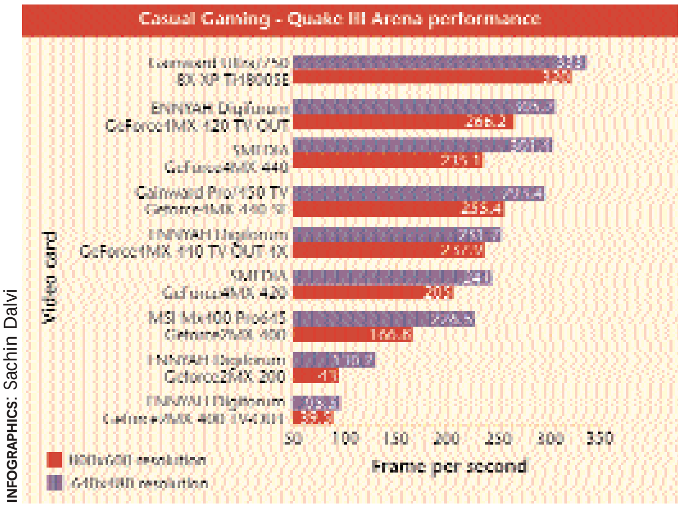
How they fared

When we talk about graphics cards, the conversation melts down to ATi and nVidia. Solutions from Matrox and Trident, although present, failed to impress us. The latest offering from these companies is nVidia's GeForce 4Ti4800 chipset and the 9700PRO from ATi. However, cards with nVidia's GeForce4MX chipsets dominated the comparison, doing very well in the casual gaming application area and performing not too shabbily in the high-end gaming sector. The reason was simple: a very few games support all the latest features that a card offers—you will often find a lag of six to 12 months between the release of a new API and that of a game that fully utilises it. It is a rule driven by economy of numbers. Thus, a card with a decent feature set and an affordable price comes across as the ideal buy. Since almost all the features of the latest and the fastest nVidia GPU are offered by the GeForce4 MX chipsets and at a relatively lower price, the result was not surprising. The high-scorers within the high-end gaming area were the latest chipsets mentioned above—another result that did not come as a surprise. The point to note, however, is that ATi solutions are now capable of taking the fight to nVidia, both in performance and in price.

Casual gaming

While a definition of a casual gamer will be hard to form, you could consider yourself one of this kind if you find yourself taking in the occasional gaming goodness, with an interest that does not exceed a few minutes. A quick bot-match in *Quake III Arena*, a mission or two in *Medal of Honor: Allied Assault*, a brief tussle with gems in *Bejeweled*, or a frag-session amongst friends in *Counter-Strike* or *Unreal Tournament*. Such activities tend to define you as a casual gamer. A video card suited to your needs will also be more than adequate for other mundane chores of browsing, word-processing, e-mailing, and satisfactorily working with CAD/3D modelling applications. Such a video card need not run the latest games at the highest resolutions and more importantly, should be affordable. No use sinking money into an activity that you only indulge occasionally. Thus, the application area of casual gaming was defined, and benchmarks run at resolutions of 640x480 and 800x600 formed the backbone of the tests for this area. On the features side, we awarded points to a card if it had Video In and Out ports, if it had a fast enough speed, a decent RAMDAC (to support those high resolutions), compliance with gaming APIs, etc. After the relevant scores were thus gathered, we calculated a Suitability to Casual Gaming Index—an STTI score that determined the success of a particular card in this application area. The logic was simple: the greater the score, the better the card. We further zoomed in on the ideal solutions area by factoring in price. Thus, all cards that scored above 50 STTI and sold at between Rs 2,500 and Rs 5,000 were considered ideally suited to the application area of casual gaming. Based on this band, we present the following analysis.



INFOGRAPHICS: Sachin Dalvi

The best video cards for Casual Gaming: The ideal frame rate that a video card should offer is around the 60 fps mark. Although the human eye is said to satisfactorily accept a 30 fps rate for moving pictures, the nature of a game calls for a buffer. A game is a dynamic creature, one moment you will be staring at a serene landscape littered with nothing but interesting trees and rocks, only to be suddenly surrounded by ghastlies that come fattened on polygons and in great numbers. Moments such as these stress a video card, which if it were hitherto running at around the 30 fps marks would now crawl at less than the ideal frame rate, causing visible stutters. Thus, a greater frame rate is always advisable—for the casual gaming area, this rate should fall within the resolutions of 640x480 and 800x600.

Two benchmarks were given greater weightage here, *Quake III Arena* and 3DMark 2001 SE. Video cards from ENNYAH dominated this area of gaming. Decent performances at very low prices elevated them to this status. The ENNYAH Digiforum GeForce2MX 200 posted scores of 130.2 fps at 640x480 and 43 fps at 800x600

10 Steps to Ensure Healthy Viewing

Before opening up the computer cabinet and replacing your old video card with a new one, here are a few things that you need to do:

STEP 1 Un-install the drivers for the old graphics card from the Add/Remove section of the Control Panel.

STEP 2 Shut down your computer and open your cabinet. Pull out the cord supplying power to your computer.

STEP 3 Remove the VGA connector attaching your monitor to your video card.

STEP 4 Carefully pull out the old graphics card, ensuring that you are electrically grounded.

STEP 5 Hunt for the AGP slot: this should be brown in colour and smaller than the similar looking white coloured PCI slots. Though unlikely, the card might be sitting on a PCI slot.

STEP 6 Insert the new graphics card into this brown slot and screw it in for a comfortable fit.

STEP 7 Close the cabinet and switch on the computer.

STEP 8 The new card will be now recognised by Microsoft Windows and the basic VGA drivers will be installed.

STEP 9 When asked to install the drivers for the card, choose No and continue.

STEP 10 Insert the driver disk provided by the card manufacturer and continue on-screen instructions to install drivers for the card. Restart machine and there you are. Your new graphics card is ready to boogie. In case your card does not come with driver software, you can download the generic Detonator drivers if your card has an nVidia video chipset (find them online at ww.nvidia.com), the CATALYST drivers if you have an ATi-chipset-based card (www.ati.com), or Matrox drivers if so required by your card (www.matrox.com).